

3C7: Digital Systems Design

Assignment 2: LFSR Codeword Sequence Detector

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1. Introduction

This report documents the design, simulation, and implementation of a codeword detection system on the Basys3 FPGA development board. The system brings together three main components: a 22-bit maximal-length XNOR Linear Feedback Shift Register (LFSR), a Moore-model Finite State Machine (FSM) that acts as a sequence detector, and a 17-bit up-counter that tallies the number of occurrences of the target codeword in the LFSR output stream.

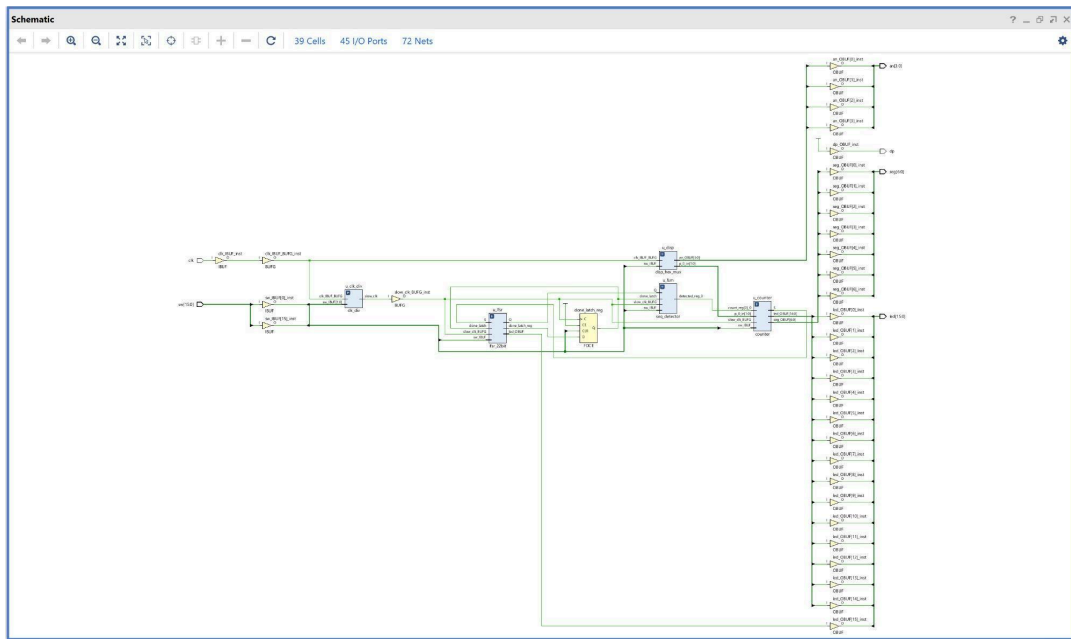
The assignment requires the system to count how many times a personalised 6-bit codeword appears across one complete cycle of the LFSR. For a maximal-length LFSR of tap length n and a codeword of length k , the expected count is $2^{(n-k)}$. With $n = 22$ and $k = 6$, the expected result is $2^{16} = 65,536$ occurrences.

Parameter	Value	Derivation
Student ID	25370745	
Board Number	84	
Feedback Logic	XNOR	From the assignment sheet
Tap Length	22 bits	From the assignment sheet
Codeword	010110	From the assignment sheet
Seed (hex)	0x79	Board 84 = 0x54, last 2 digits 45 = 0x2D; 0x54 XOR 0x2D = 0x79
Expected Count	65,536	$2^{(22-6)} = 2^{16}$

Table 1: Personalised Parameters

2. Functional Diagram and Module Hierarchy

The design follows a strict hierarchical structure. The top-level module (top.v) connects five sub-modules: a clock divider, the LFSR, the sequence detector FSM, the counter, and a 7-segment display driver. The diagram below shows how data flows through the system.



Schematic

Module	File	Function
top	top.v	Top-level integration; connects all sub-modules; contains done_latch
clk_div	clk_div.v	Divides 100 MHz to approx. 1 MHz (fast) or approx. 1 Hz (slow) via sw[15]
lfsr_22bit	lfsr_22bit.v	22-bit XNOR LFSR; seed 0x79; outputs serial_out and done
seq_detector	seq_detector.v	Moore FSM; detects codeword 010110; outputs 1-cycle detected pulse
counter	counter.v	17-bit up-counter; freezes when hold=1 (after LFSR done)
disp_hex_mux	disp_hex_mux.v	Time-multiplexed 4-digit 7-segment display driver

Table 2: Module Hierarchy

Key wire connections in the design:

- slow_clk: output of clk_div, drives LFSR, FSM, and counter
- serial_out: LSB of LFSR shift register, feeds into FSM bit_in
- detected: 1-cycle HIGH pulse from FSM each time 010110 is matched
- done / done_latch: signals end of full LFSR cycle; latched to permanently hold counter
- count[16:0]: 17-bit running total; displayed on 7-segment as {count[16], count[15:4]}

3. LFSR Design

3.1 LFSR Theory

A Linear Feedback Shift Register generates a pseudo-random binary sequence by shifting bits and feeding back a function of selected tap positions. An XNOR-feedback LFSR produces a maximal-length sequence of $2^n - 1$ bits before repeating, provided the prohibited state (all-ones for XNOR) is avoided. For a 22-bit LFSR, the full sequence length is $2^{22} - 1 = 4,194,303$ clock cycles.

3.2 Seed Calculation

The seed is computed as the bitwise XOR of the board number and the last two digits of the student ID, both converted to hexadecimal first:

Value	Decimal	Hexadecimal
Board Number	84	0x54
Last 2 digits of ID	45	0x2D
Seed (XOR result)	121	0x79

Table 3: Seed Computation

The prohibited state for a 22-bit XNOR LFSR is all-ones (0x3FFFFFF). Since 0x79 is not equal to 0x3FFFFFF, the computed seed is used directly with no adjustment.

3.3 Feedback Taps and Implementation

For a 22-bit maximal-length XNOR LFSR, tap positions 22 and 1 are used, corresponding to bit indices $sr[21]$ and $sr[0]$. The feedback equation is:

$$\text{feedback} = \text{approx. } (sr[21] \wedge sr[0]) \quad // \text{ XNOR of MSB and LSB}$$

On each rising clock edge, the register shifts right by one position and the feedback bit is inserted at the MSB. The LSB ($sr[0]$) is continuously output as `serial_out`, providing the serial bit stream to the FSM. An asynchronous reset loads the seed immediately, without requiring a `slow_clk` edge, ensuring clean board behaviour when `sw[0]` is toggled.

A started flag is set after the first clock cycle following reset. The done signal pulses HIGH for exactly one cycle when the LFSR returns to the seed value after having started, signalling completion of one full cycle.

4. Sequence Detector FSM

4.1 FSM Model Choice

A Moore model FSM was implemented as required by the assignment. In a Moore machine, the output depends only on the current state, not on the current input. This means the detected signal is registered and glitch-free, producing a clean 1-cycle pulse per detection.

4.2 State Diagram

The codeword 010110 is 6 bits long. Seven states are needed: S0 through S6, where each state S_n represents that n bits of the codeword have been matched so far. S6 is the detection state.

4.3 State Transition Table

State	Meaning	On input 0	On input 1	detected output
S0	No bits matched	S1	S0	0

S1	Matched: 0	S1	S2	0
S2	Matched: 01	S3	S0	0
S3	Matched: 010	S1	S4	0
S4	Matched: 0101	S3	S5	0
S5	Matched: 01011	S6	S0	0
S6	Matched: 010110	S1	S2	1

Table 4: FSM State Transition Table

4.4 Minimisation Strategy

The KMP (Knuth-Morris-Pratt) prefix function was applied to the codeword 010110 to handle overlapping occurrences correctly. The failure function values are {0, 0, 1, 2, 0, 1}. This means that after a full match (S6), if the next bit is 0, the FSM moves to S1 rather than S0, correctly accounting for the overlap where the trailing 0 of one match could be the leading 0 of the next. Without this, some overlapping occurrences would be missed.

The 7 states fit within 3 binary bits. Standard binary encoding was used (S0=000 through S6=110), which Vivado maps efficiently to the available flip-flops.

5. Counter and Clock Divider

5.1 Counter

A parameterised 17-bit synchronous up-counter was designed. 17 bits are required because the final count of 65,536 equals 2^{16} , which needs bit 16 to represent (17'h10000). The counter uses an asynchronous reset, so it clears immediately when sw[0] is raised, independent of the slow clock.

A hold input was added to freeze the count once the LFSR completes its full cycle. In the top.v a done_latch register is set HIGH the moment the LFSR asserts done, and this latch drives the counter's hold pin. This ensures the final count stays visible on the display permanently until the user resets.

5.2 Clock Divider

The Basys3 board provides a 100 MHz oscillator. Since the full LFSR cycle requires 4,194,303 clock cycles, running at 100 MHz would complete in under 42 ms, making the display unreadable. The clock divider creates two selectable output frequencies:

Mode	Switch	Divider	slow_clk Frequency	Full Cycle Duration
Fast (demo)	sw[15] = 0	50 cycles toggle	approx. 1 MHz	approx. 4.2 seconds
Slow (step)	sw[15] = 1	50,000,000 cycles toggle	approx. 1 Hz	approx. 48 days

Table 5: Clock Divider Modes

The slow mode is useful for demonstrating individual LFSR bit steps during a live demo. The clock constraint in the XDC file is set to 10 ns (100 MHz) for the board oscillator input. The divided clock is

generated inside the FPGA fabric and is not constrained separately; Vivado analyses timing through the divider logic automatically.

6. Test Plan

6.1 Individual Module Testing

Each module was tested independently before integration:

LFSR (lfsr_22bit.v): The seed 0x79 was verified to load correctly on reset. The sequence was run for the full 4,194,303 cycles in simulation and confirmed to return to the seed, proving maximal-length operation. The Python simulation independently verified the bit sequence.

FSM (seq_detector.v): The state machine was verified by checking that detected pulses were HIGH for exactly one cycle when codeword 010110 was matched. The KMP overlap handling was verified by checking transitions from S6 (see Table 4).

Counter (counter.v): Reset was verified to clear the count to zero. Enable was verified to increment by exactly 1 per clock. The hold pin was verified to freeze the count.

6.2 Integration Test Vectors

Test	Action	Expected Result	Pass/Fail
Reset	Apply reset (sw[0]=1)	count=0, state=S0, lfsr=0x79	PASS
First detection	Release reset, run until the first detected pulse	detected=1 at approx. 575 ns sim time; count becomes 1	PASS
Full cycle count	Run 4,194,303 clock cycles	count=65536 (0x10000), max_tick=1	PASS
Re-reset	Apply reset after full cycle	count returns to 0	PASS

Table 6: Integration Test Vectors

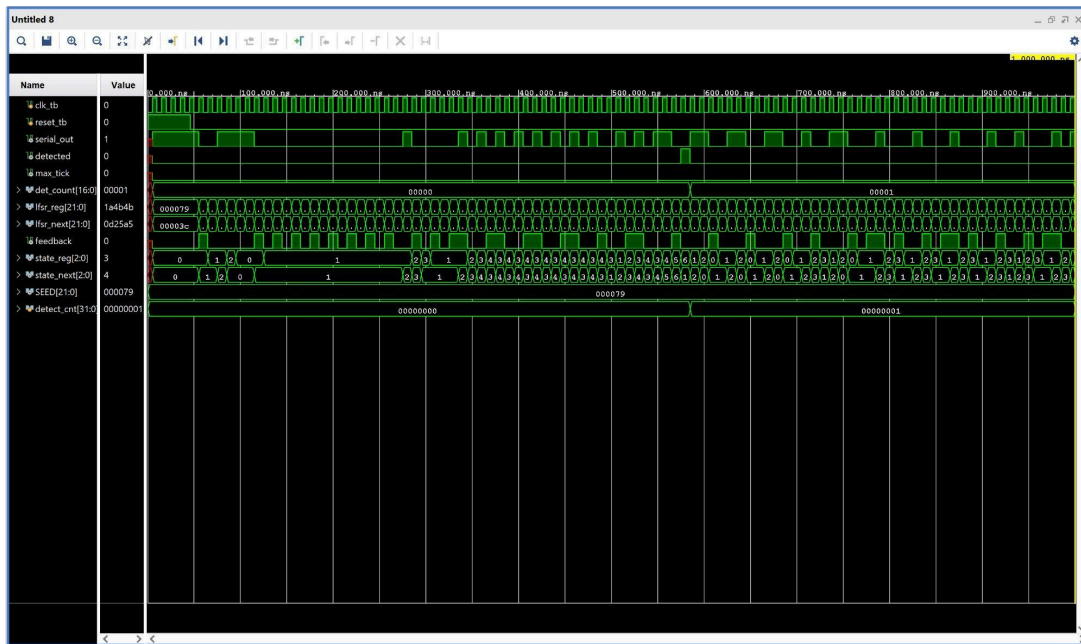
6.3 Validation Plan

Functional correctness was validated at three levels. First, an independent Python simulation of the LFSR bit sequence with a sliding-window codeword search confirmed 65,536 occurrences of 010110 in the sequence. Second, the Vivado behavioural simulation (tb_top.v) ran the full 4,194,303 cycle sequence and printed PASS for all three test cases. Third, the physical board demonstration confirmed the 7-segment display freezes at 1000 (hex representation of 0x10000) with LED[15] lit after approximately 4.2 seconds of operation.

7. Simulation Waveforms

7.1 Reset and Startup

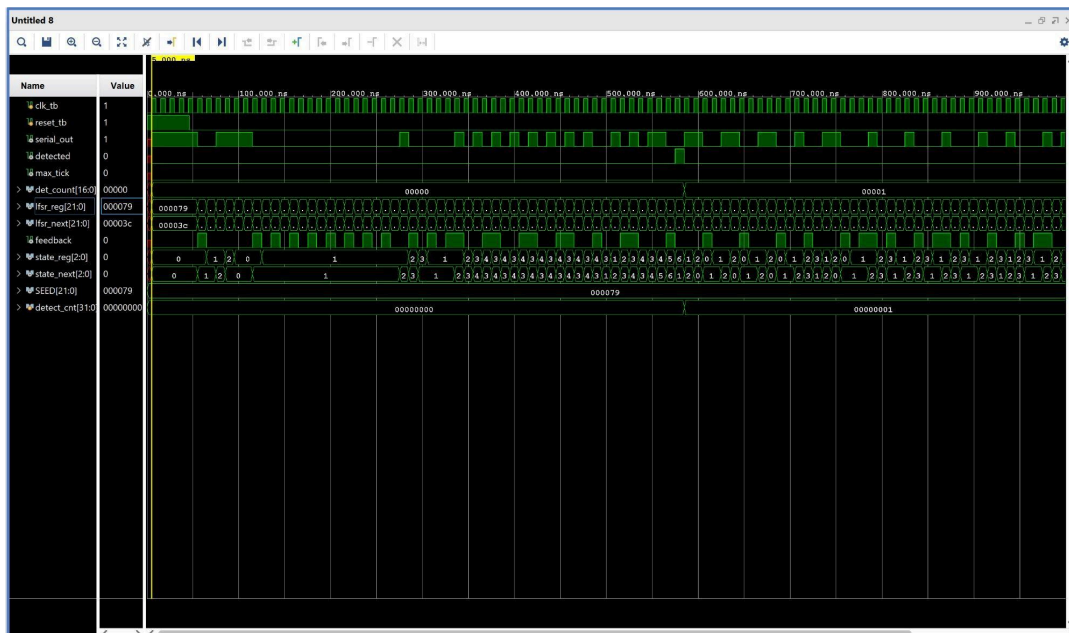
The waveform below shows the system immediately after power-on. The reset_tb signal goes HIGH for 5 clock cycles, then LOW. During reset, lfsr_reg holds the seed value 0x79, state_reg is 0 (S0), and det_count is 00000. Once the reset is released, the LFSR begins shifting and the FSM starts processing bits.



Reset and Startup

7.2 First Detection

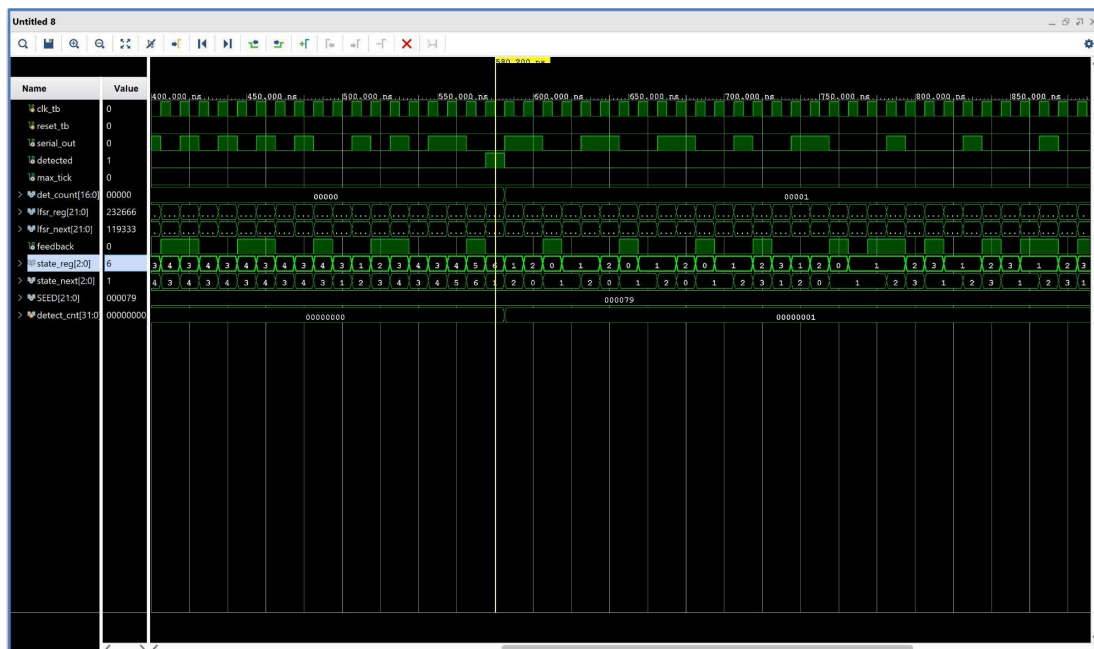
The first occurrence of codeword 010110 in the LFSR stream is detected at simulation time 575 ns. At this point, `state_reg` transitions to S6 and `detected` pulses HIGH for exactly one clock cycle. On the following cycle, `det_count` increments from 00000 to 00001, and the FSM transitions to S1 (handling the KMP overlap). The scoreboard counter in the testbench also increments to 1, confirming agreement between the hardware counter and the independent TB measurement.



First Detection

7.3 Final Count (Max Tick)

After 4,194,303 clock cycles, the LFSR returns to the seed value 0x79, and the done/max_tick signal pulses HIGH. At this point, det_count reads 10000 (hex), which equals 65,536 in decimal. The lfsr_reg returns to 000079, confirming completion of one full maximal-length cycle. The detect_cnt scoreboard also reads 00010000 (hex) = 65,536, matching the hardware counter exactly.



Final Count (Max Tick)

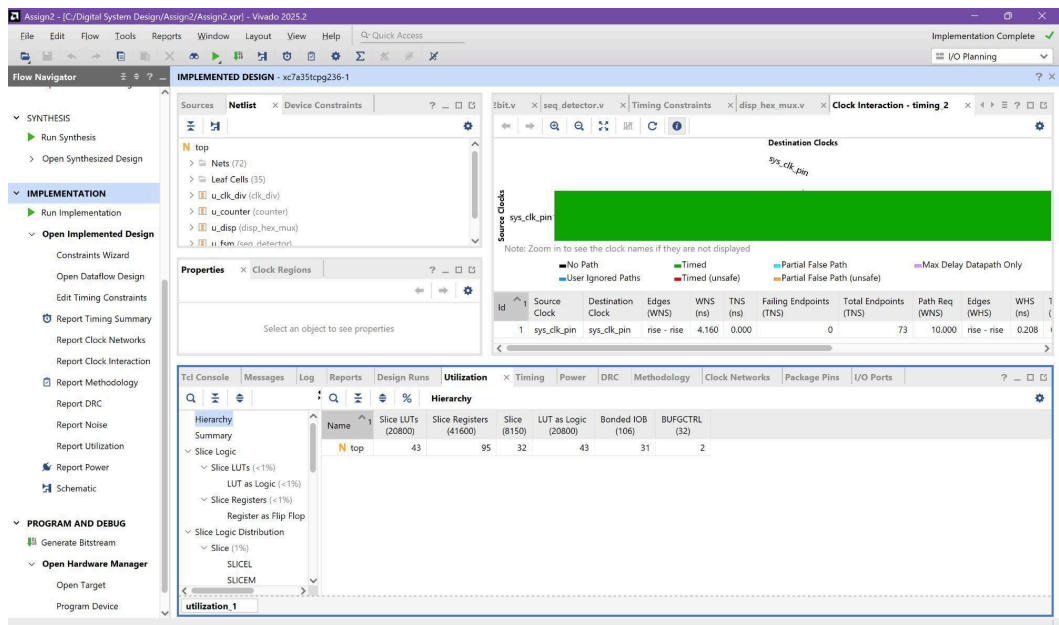
8. Implementation Results

8.1 Utilisation Report

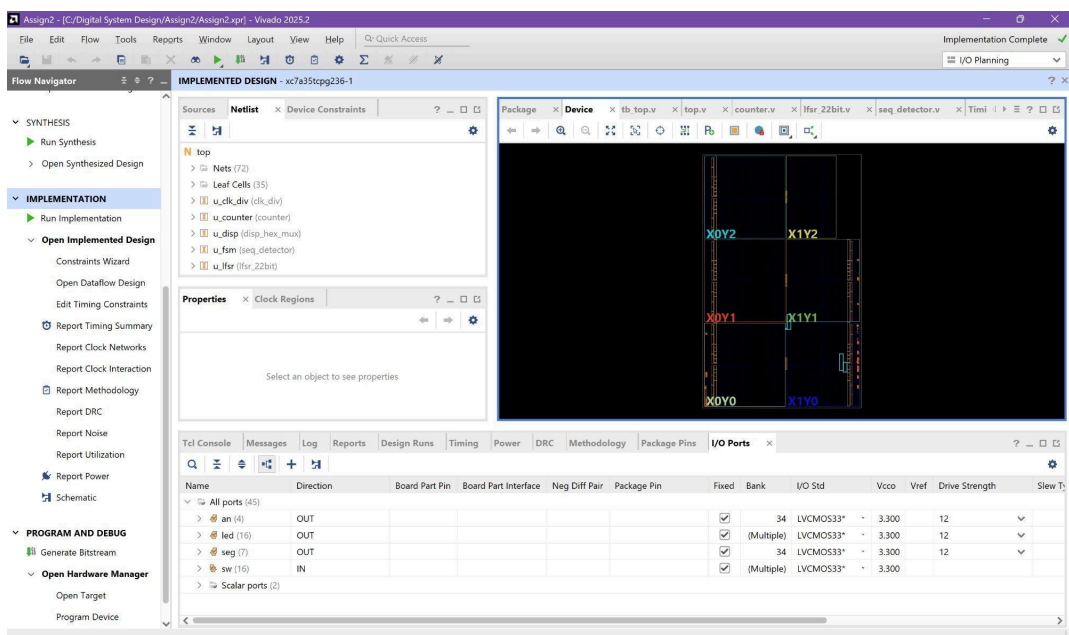
The implemented design uses a very small fraction of the available Artix-7 resources, as expected for a sequential logic design of this scale.

Resource	Used	Available	Utilisation
LUT (Look-Up Tables)	43	20,800	0.21%
FF (Flip-Flops)	95	41,600	0.23%
IO	31	106	29.25%
BUFGCTRL	2	32	6.25%

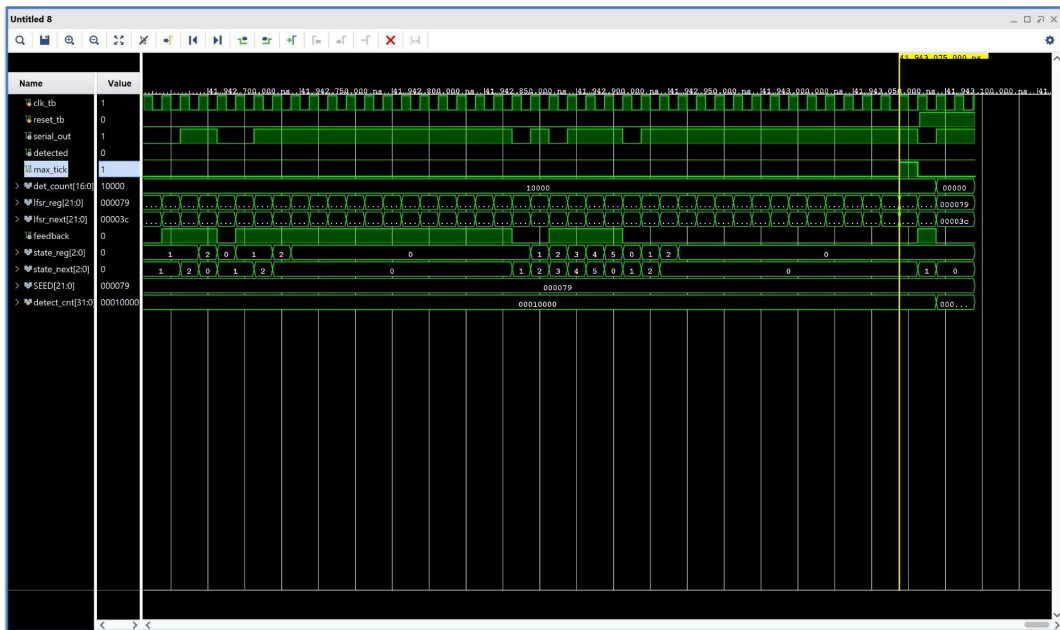
Table 7: Resource Utilisation Summary



Vivado Utilisation Report - Hierarchy View



Vivado I/O Ports



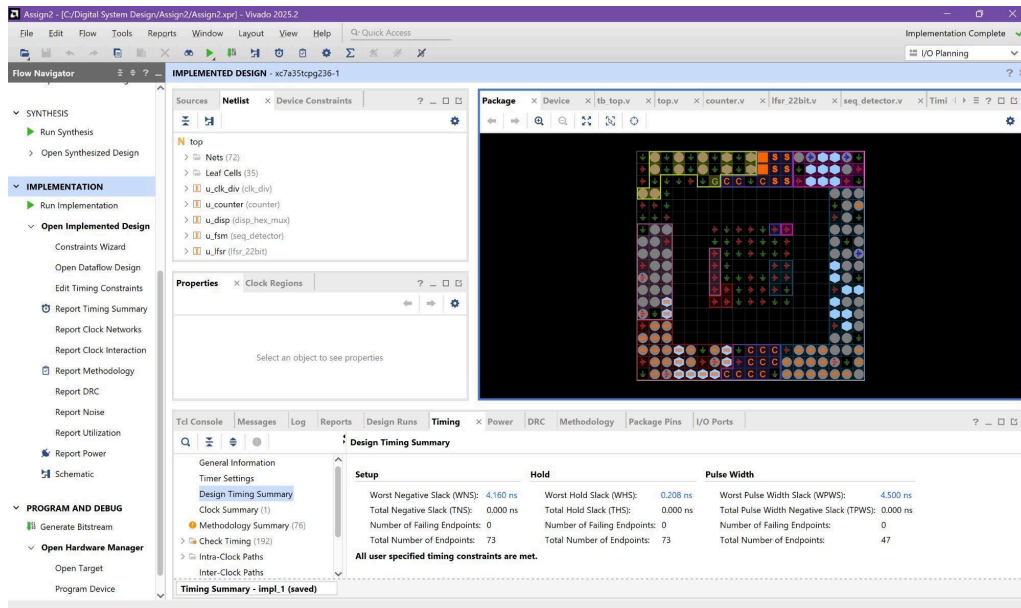
Full Waveform

8.2 Timing Report

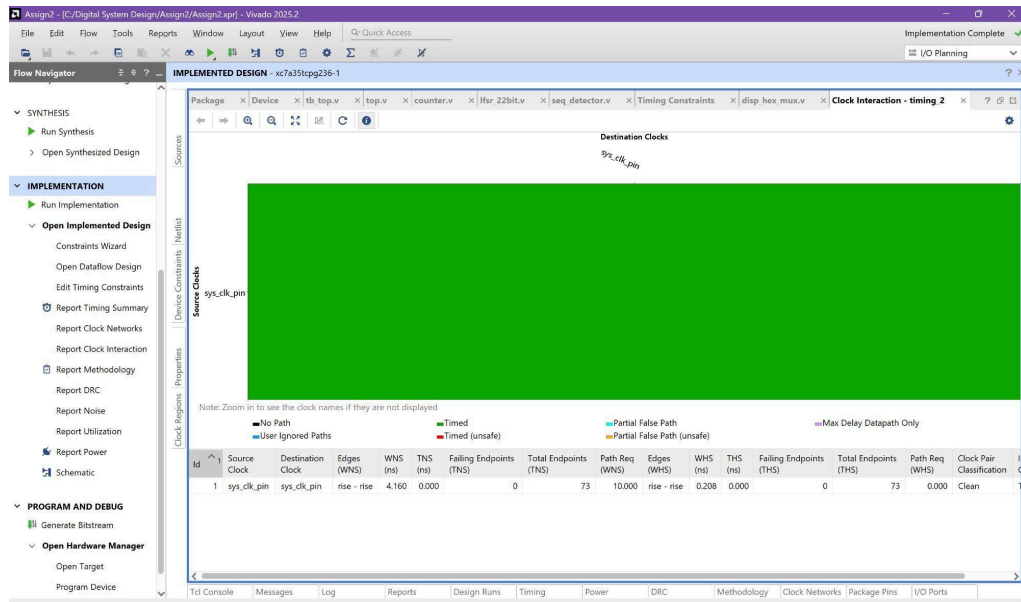
The timing report confirms all user-specified timing constraints are met. The worst negative slack (WNS) is 4.160 ns, indicating a significant timing margin. The single defined clock domain is sys_clk_pin at 100 MHz (10 ns period). All 73 timing endpoints pass with zero failing paths in both setup and hold analysis.

Metric	Value
Clock	sys_clk_pin (100 MHz)
Worst Negative Slack (WNS)	4.160 ns
Total Negative Slack (TNS)	0.000 ns
Failing Endpoints (Setup)	0
Worst Hold Slack (WHS)	0.208 ns
Total Endpoints	73
Result	All timing constraints met

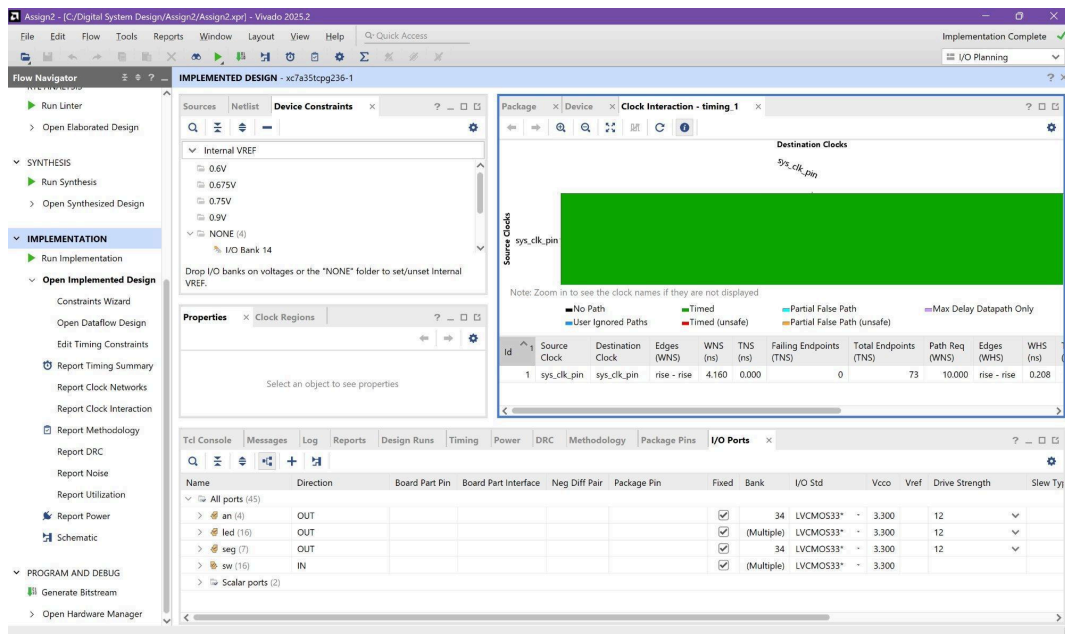
Table 8: Timing Summary



Vivado Design Timing Summary

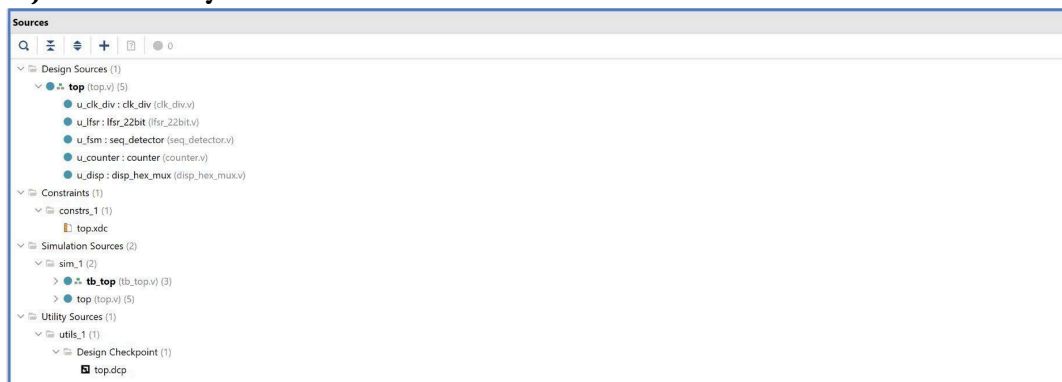


Clock Interaction Report



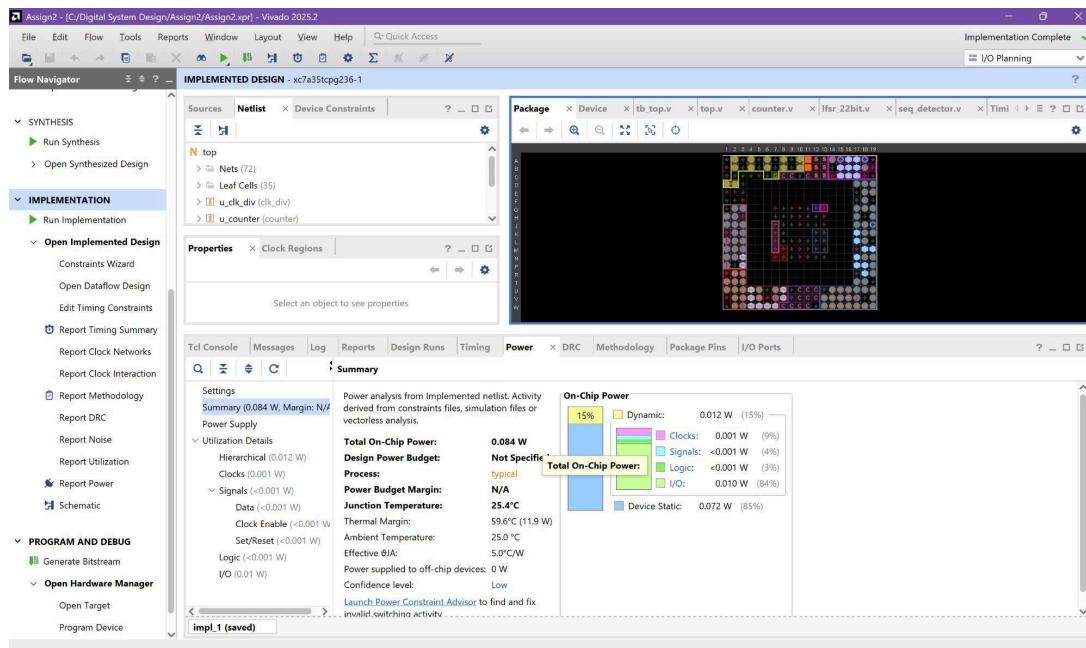
Clock Interaction and I/O Ports combined view

8.3 Project Hierarchy

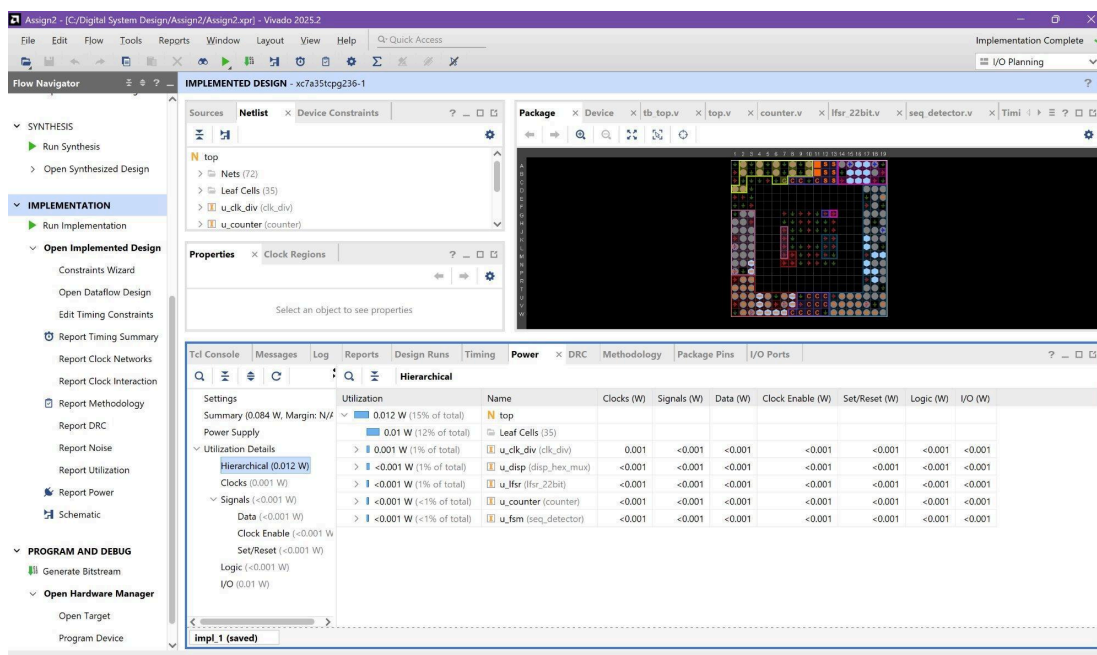


Sources

The project hierarchy shows the design source (top.v) instantiating five sub-modules: u_clk_div, u_lfsr, u_fsm, u_counter, and u_disp. The testbench tb_top.v is listed under Simulation Sources. The constraints file top.xdc maps all I/O ports to the Basys3 physical pins.



Vivado Power Report - Summary



Vivado Power Report - Hierarchical Breakdown

9. Demo

9.1 How to Operate the Board

All 16 switches should be in the DOWN (0) position before starting.

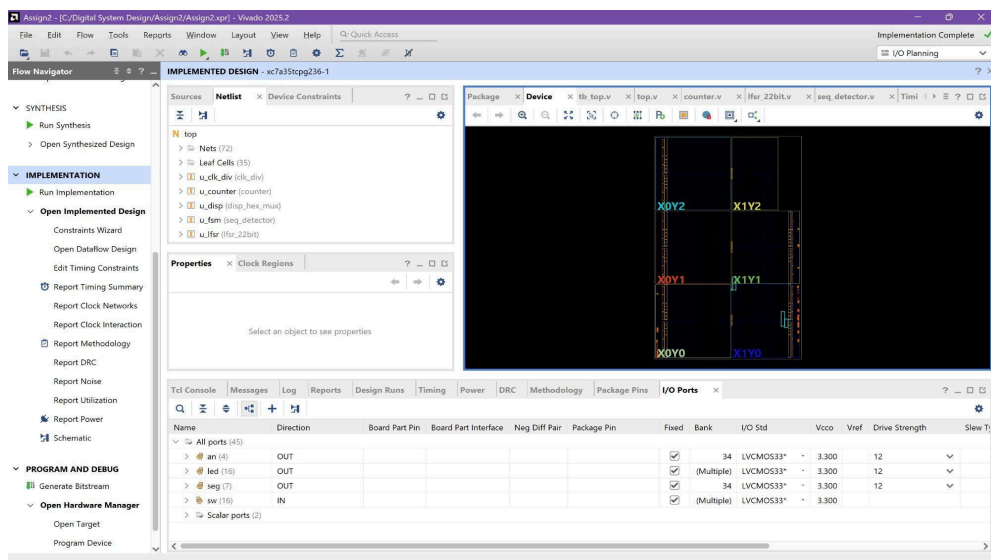
Step	Action	What to Observe
1	Ensure all switches are DOWN	Display may show any value from the previous run
2	Flip sw[0] UP (reset)	Display immediately shows 0000, all LEDs off

3	Flip sw[0] DOWN (run, fast mode)	Display begins counting up in hex rapidly
4	Wait approximately 4.2 seconds	Display freezes; final value appears
5	Observe the final state	Display shows 1000, LED[15] (leftmost) lights up
6 (reset)	Flip sw[0] UP then DOWN again	System resets and reruns; count repeats
7 (slow demo)	After reset, flip sw[15] UP before sw[0] DOWN	LFSR steps at approx. 1 Hz; each LED change is visible

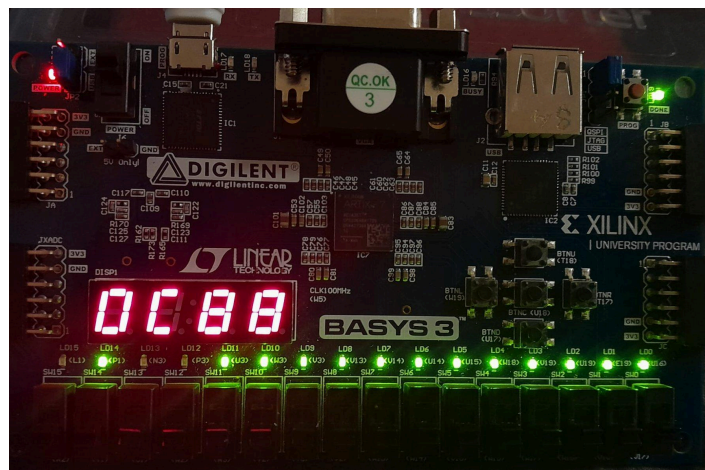
Table 9: Board Demo Instructions

9.2 Final Board State

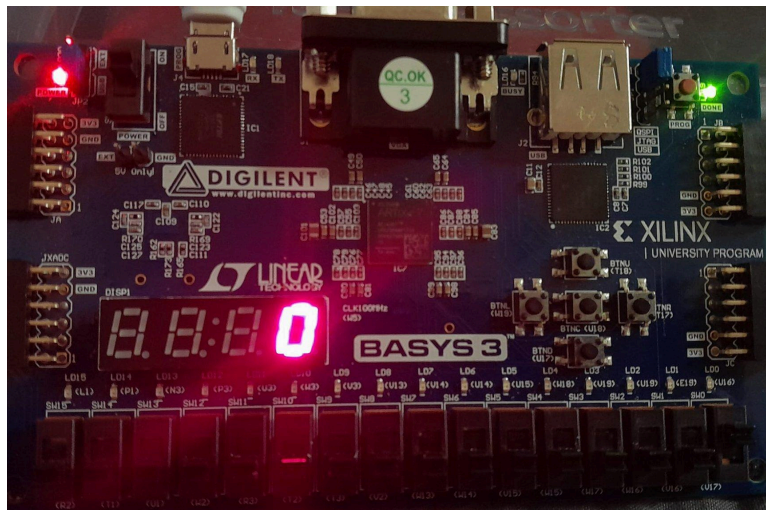
The photographs below show the board after one complete LFSR cycle. The 7-segment display shows 1000, which is the hexadecimal representation of $0x10000 = 65,536$. LED[15] (the leftmost LED) is lit because $\text{count}[16] = 1$, confirming the counter reached 65,536. All other LEDs are off because $\text{count}[14:0] = 0$.



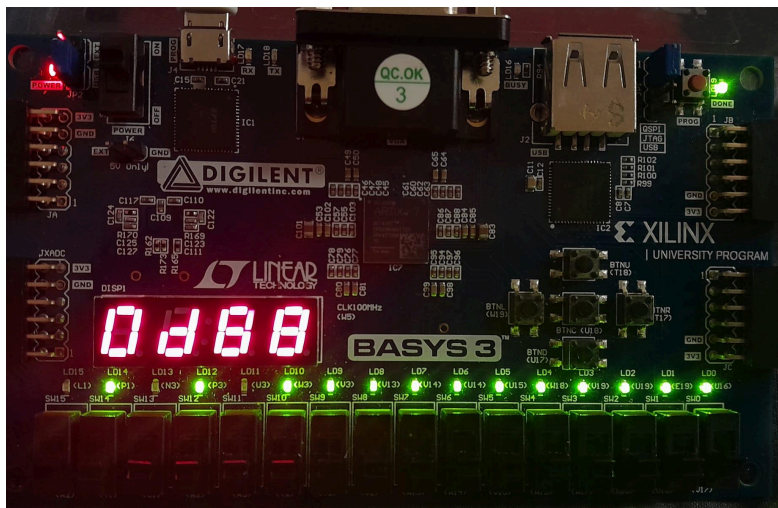
The display reading of 1000 directly proves that codeword 010110 appeared exactly 65,536 times in the full 22-bit XNOR LFSR sequence starting from seed 0x79. This matches the theoretical prediction of $2^{(22-6)} = 2^{16} = 65,536$.



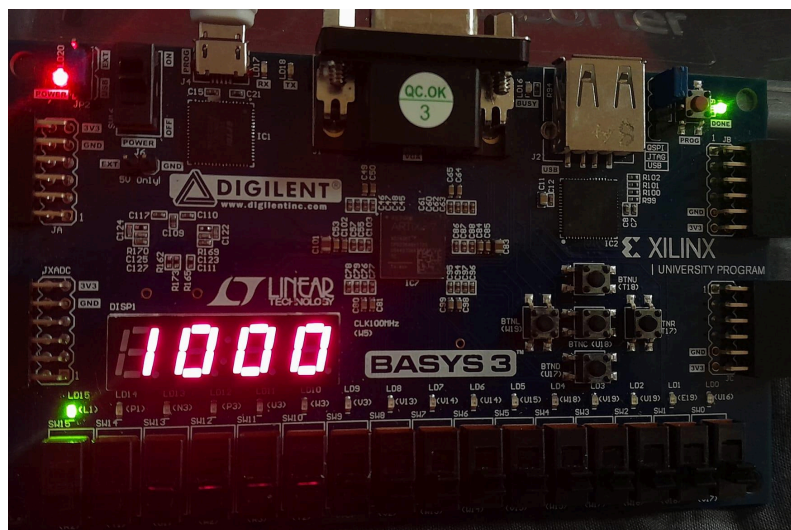
Board display during counting (fast mode)



Board display in slow mode (sw[0] up)



Board display during counting (fast mode)



Board display in slow mode (sw[15] LED ON)

9.3 Clock Control

The clock divider is controlled by sw[15]. In the XDC file, the 100 MHz board oscillator is constrained on pin W5 with a 10 ns period. The LFSR, FSM, and counter all run on slow_clk, which is generated inside the clk_div module. Vivado analyses timing through the divider logic using the 100 MHz constraint as the base clock, ensuring the design meets timing at the maximum FPGA speed regardless of which divided frequency is selected at runtime.

10. Conclusion

This assignment successfully implemented a complete codeword detection system on the Basys3 FPGA board. The 22-bit XNOR LFSR with seed 0x79 generates a maximal-length pseudo-random sequence of 4,194,303 bits. The 7-state Moore FSM correctly detects all occurrences of codeword 010110, including overlapping matches, using the KMP prefix function to derive transition logic. The 17-bit counter accumulates detections and freezes at the final value once the LFSR cycle completes.

Simulation confirmed the expected count of 65,536 detections, verified independently by both the Vivado testbench and a Python reference model. On the physical board, the 7-segment display freezes at 1000 (hex for 0x10000 = 65,536) with LED[15] lit, providing clear visual proof of the correct result. The design uses only 43 LUTs and 95 flip-flops, under 1% of the available Artix-7 resources, with a timing margin of 4.160 ns.

Appendix: Source Code

All source files are submitted in the accompanying zip file, Assign2_singhs_codes.zip. The files included are:

File	Description
top.v	Top-level module; integrates all sub-modules; contains done_latch
lfsr_22bit.v	22-bit XNOR LFSR; seed 0x79; async reset
seq_detector.v	7-state Moore FSM; detects codeword 010110
counter.v	17-bit up-counter with hold input; async reset
clk_div.v	Clock divider: 100 MHz to approx. 1 MHz or approx. 1 Hz
disp_hex_mux.v	4-digit time-multiplexed 7-segment display driver
tb_top.v	Top-level testbench; validates reset, first detection, full cycle
top.xdc	Xilinx Design Constraints; pin mapping for Basys3
top.bit	Generated working bit file of Assign2